

MS&E 228 (Winter 2026) — Lecture 6 Notes (Student Handout)

The Doubly Robust Estimator: Efficiency and Application

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1 Introduction

In our previous lectures, we have systematically built a powerful and versatile toolbox for estimating the Average Treatment Effect (ATE) from observational data. Our journey began with simple methods like linear regression, progressed to more flexible machine learning-based approaches for modeling the outcome (the G-formula) and treatment assignment (Inverse Propensity Weighting), and culminated in the state-of-the-art **Doubly Robust (DR)** estimator.

The DR estimator represents the pinnacle of this line of research, combining the strengths of both the G-formula and IPW. Its defining feature is its *double robustness*: it provides a consistent estimate of the ATE if either our outcome regression model $\hat{g}(D, X)$ or our propensity score model $\hat{p}(X)$ is consistently estimated. This property provides a crucial safety net against model misspecification, making it the preferred method for many applied causal inference problems.

This lecture is dedicated to a deeper exploration of the DR estimator. Having established *how* to use it, we now turn to *why* it is so powerful and *how* it behaves in different settings. We will structure our investigation around two key themes:

1. **The DR Estimator in Practice:** We will move from abstract concepts to concrete applications, examining how the DR estimator is used to solve real-world challenges in a clinical trial setting and how it is implemented in major industry software packages used by companies like Microsoft and Uber.
2. **The Statistical Efficiency of the DR Estimator:** We will conduct a formal analysis of the estimator's variance. This will lead us to the concept of the *semiparametric efficiency bound*, a theoretical benchmark for precision that the DR estimator achieves. This analysis will not only solidify our understanding of the DR estimator's properties in observational studies but will also lay the essential groundwork for the next chapter, where we will explore how to best use covariates for variance reduction in the simpler setting of a Randomized Controlled Trial (RCT).

2 Looking-Back: Estimator Pros and Cons

As part of an in-class activity, we discussed the pros and cons of all the types of estimators we have learned thus far. This exercise is crucial for developing practical intuition about which tool to use in which scenario. A summary of our discussion is presented below, with the full table included in the Appendix.

In-Class Activity Summary: Estimator Pros and Cons

- **Linear Regression:** Simple, interpretable, and provides confidence intervals. Its simplicity means no hyperparameter tuning, which reduces the risk of *p-hacking*. However, it is likely misspecified if the true Conditional Expectation Function (CEF) is not linear, and it can extrapolate poorly if there is a lack of overlap in covariate distributions between treatment groups.
- **G-formula (ML-based):** Can capture non-linearities. However, it can be slow to converge statistically and does not provide straightforward confidence intervals. A key distinction arises between S-learners and T-learners. The **S-learner** may regularize away the treatment effect if it is small, leading to biased estimates. The **T-learner** avoids this by fitting separate models, but can perform poorly with imbalanced treatment groups due to reduced sample size for the smaller group.
- **IPW (ML-based):** Only requires modeling the treatment assignment mechanism, which can be advantageous if the outcome model is complex. However, it can have very high variance, especially with poor overlap (extreme propensities), and it is not straightforward to compute confidence intervals when using flexible ML models.
- **Doubly Robust (DR):** The state-of-the-art. It is consistent if either the outcome or propensity model is correct, has lower variance than IPW, and provides valid confidence intervals even when using flexible ML models for estimation. Its main drawbacks are that its theoretical guarantees rely on conditions (like the product rate condition) that cannot be empirically verified, and its estimates have an extra element of variability due to the randomness of k-fold cross-fitting.

2.1 A Deeper Look: S-Learner vs. T-Learner Trade-offs

A particularly important distinction emerged from our discussion regarding the G-formula: the difference between the S-learner and the T-learner. This is a subtle but consequential choice that affects both the bias and variance of the resulting estimator.

The Regularization Problem with S-Learners. When fitting an S-learner, the model optimizes for the overall mean squared error of predicting the outcome Y from both the treatment D and the covariates X . If the treatment effect is small relative to the predictive power of the covariates, the regularization process may effectively “throw away” the treatment variable. Consider fitting a depth-two decision tree as an S-learner. The algorithm will choose the two most predictive features for its splits. If covariates X_1 and X_2 are highly predictive of Y , the

tree will likely use them for its splits and ignore the treatment variable D entirely. The result is a model that predicts $\hat{g}(1, X) = \hat{g}(0, X)$ for all X , leading to a treatment effect estimate of exactly zero—with high confidence.

The T-Learner Solution. The T-learner avoids this problem by fitting separate models for the treated and control groups. This is equivalent to forcing the treatment variable to be the first split in a decision tree. The treatment effect can never be regularized away. However, this comes at a cost: the T-learner does not share information between the two groups. If the treatment group is small (e.g., 5% of the sample), the model for the treated group is trained on very little data, which can lead to high variance or poor accuracy.

Practical Recommendation. In practice, if the dataset is not severely imbalanced, the T-learner is often the preferred choice due to its better bias properties. A potential middle ground is to use an S-learner but configure the model to not regularize the treatment variable, if the model architecture allows for it.

3 The DR Estimator in Practice

To motivate our deep dive into the theory, let's first ground our discussion in how the DR estimator is applied to solve real-world problems, from academic research to large-scale industry applications.

3.1 Case Study: Estimating Per-Protocol Effects in the EAGeR Trial

A common and significant challenge in clinical trials is participant non-adherence. While the primary analysis of an RCT often focuses on the Intention-to-Treat (ITT) effect—the effect of being *assigned* to treatment—researchers and stakeholders are often more interested in the Per-Protocol effect—the effect of actually *adhering* to the prescribed treatment.

3.1.1 ITT vs. Per-Protocol Analysis

It is important to understand the distinction between these two types of analysis:

- **Intention-to-Treat (ITT):** This measures the effect of being *assigned* to a treatment group, regardless of whether the participant actually adhered to the treatment. ITT analysis preserves the benefits of randomization and is often the primary analysis for regulatory purposes. However, it can be diluted by non-adherence.
- **Per-Protocol Effect:** This measures the effect of actually *adhering* to the treatment. This is often the scientific question of greater interest, but it breaks the randomization, because adherence is a choice and can be confounded by other factors.

The Challenge: Per-protocol analysis of an RCT must be treated as an **observational study**, because adherence can be confounded by baseline and post-randomization factors. The reason someone adheres to the treatment protocol may be related to the same factors that influence their outcome. For instance, a patient who is feeling sicker may be more motivated to take their medication, but their underlying condition also makes them more likely to have a poor outcome. This creates a confounding relationship that breaks the randomization.

The ITT Effect as a Diluted Estimate. It is useful to think of the ITT effect as a product of two quantities: the probability of adhering to the treatment and the effect of the treatment itself. If adherence is only 50%, the ITT estimate will be roughly half of the true per-protocol effect. This is why the per-protocol analysis often reveals a larger effect.

3.1.2 The EAGeR Trial

The paper by Zhong et al. (2022) on the Effects of Aspirin in Gestation and Reproduction (EAGeR) trial provides a compelling example of how to tackle this challenge. The EAGeR trial was a multicenter, block-randomized, double-blind, placebo-controlled clinical trial with the following characteristics:

- **Participants:** 1,227 women with a history of pregnancy loss.
- **Intervention:** Daily low-dose aspirin (81 mg) vs. placebo.
- **Outcome:** hCG-detected pregnancy.
- **Follow-up:** Up to 6 menstrual cycles for attempted pregnancy.

The researchers used a doubly robust estimator to control for confounding. It is worth noting that the DR estimator is known by several names; in the biostatistics literature, it is often called the **Augmented Inverse Propensity Weighted (AIPW)** estimator, while in economics it may be referred to as the DML estimator for the interactive regression model. For their implementation, the authors used the **SuperLearner** package in R, an AutoML tool that is an analogue to the **FLAML** library we have used. It performs stacking over an ensemble of models, including GLM, MARS, Random Forest, and XGBoost. The analysis was conducted with pure nested cross-validation to provide strong guardrails against overfitting, a prudent choice given the relatively small sample size of approximately 1,200 participants.

Table 1: Key Results from the EAGeR Trial Analysis (Zhong et al., 2022)

Analysis	Effect Estimate (per 100 women)	95% CI
Intention-to-Treat (ITT)	+4.3	(−1.1, 9.6)
Per-Protocol (DR with ML)	+8.0	(2.5, 13.6)

Key Insight: The per-protocol effect is nearly twice as large as the ITT effect and is statistically significant, while the ITT effect is not. This demonstrates two important points. First, adherence matters: the true biological effect of the treatment can be substantially larger than what a simple ITT analysis reveals. Second, flexible ML-based DR methods can uncover these effects by properly controlling for the confounders that drive adherence. The supplementary material of the paper provides the full R code for their analysis, serving as a useful template for implementing these methods.

3.2 Industry Packages and Large-Scale Application

The advanced methods we are studying are no longer confined to academic journals. They are implemented in robust, well-documented, open-source packages and are used in production at

some of the world's largest tech companies.

3.2.1 EconML (Microsoft)

Microsoft Research's EconML package provides a comprehensive suite of state-of-the-art causal inference estimators. The `LinearDRLearner` class implements the doubly robust approach.

```
from econml.dr import LinearDRLearner
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier

# S-learner variant by default
est = LinearDRLearner(
    model_propensity=RandomForestClassifier(),
    model_regression=RandomForestRegressor()
)
est.fit(Y, T, X=None, W=X)

# Get ATE and confidence intervals
est.summary()
```

Note: The EconML package uses an S-learner implementation by default. A T-learner can be emulated by passing it a regression model that internally fits two separate models, for which the package provides a wrapper.

```
from econml.utilities import SeparateModel

# T-learner variant
est = LinearDRLearner(
    model_propensity=RandomForestClassifier(),
    model_regression=SeparateModel(RandomForestRegressor(), RandomForestRegressor())
)
```

3.2.2 DoubleML (DoubleML Package)

The DoubleML package provides a very clean implementation of the double machine learning framework.

```
from doubleml import DoubleMLIRM, DoubleMLData
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier

# Prepare data
dml_data = DoubleMLData(df, y_col='Y', d_cols='D', x_cols=X_cols)

# T-learner API: separate models for  $E[Y|X, D=d]$ 
ml_g = RandomForestRegressor()
ml_m = RandomForestClassifier()

dml_irm = DoubleMLIRM(dml_data, ml_g, ml_m, score='ATE')
dml_irm.fit()

print(dml_irm.summary)
ci = dml_irm.confint(level=0.95)
```

Note: In contrast to EconML, the DoubleML package uses a T-learner implementation by default. When an outcome model is passed, it will be used to fit separate models for the treated and control groups.

3.2.3 Industry Application: Uber

Uber’s causal inference platform makes extensive use of doubly robust estimators for analyzing observational data at a massive scale. As detailed in a company blog post on the topic,¹ these methods are applied across various domains to understand complex treatment effects among millions of users, where computational efficiency and robustness to model misspecification are critical concerns.

Takeaway: These methods are not just academic exercises—they are deployed in production at major tech companies and are used by empirical researchers in a variety of fields.

When Should You *Not* Use the Doubly Robust Estimator?

A natural question is: when should you *not* use the DR estimator and instead opt for a simpler method like IPW or the G-formula? In most cases, the DR estimator is the recommended default. It provides confidence intervals (unlike ML-based IPW or G-formula), has lower variance than IPW, and is robust to misspecification of one of the nuisance models. However, if you have very poor overlap and are willing to make stronger parametric assumptions about the outcome model, you may achieve lower variance with a semi-parametric outcome modeling approach, as opposed to a fully non-parametric outcome model. We will explore such methods in subsequent lectures, under the topic of *partially linear models*.

4 The Statistical Efficiency of the Doubly Robust Estimator

We have motivated the DR estimator from the perspective of its double robustness and its practical application. Now, we turn to its other key property: **statistical efficiency**. Not only is the DR estimator robust to model misspecification, but it is also, in a very formal sense, optimally precise.

The Semiparametric Efficiency Bound

In many standard causal inference settings (both observational and experimental), the doubly robust estimator achieves the **semiparametric efficiency bound**. This is a theoretical lower bound on the asymptotic variance of any regular, asymptotically unbiased estimator. In simple terms, it means that **no other estimator can be more precise** in large samples without making stronger, and potentially incorrect, assumptions about the underlying data-generating process (e.g., assuming the CEF is linear).

Why this matters:

- DR is not just robust (double robustness), it’s also **optimally precise**.
- If you’re willing to make stronger assumptions (e.g., linear CEF), you can do better (e.g., OLS has lower variance under linearity).

¹See the blog post: Using Causal Inference to Improve the Uber User Experience

4.1 A Detailed Derivation of the DR Variance

To understand what drives the precision of the DR estimator, we need to unpack its variance formula. Recall the DR score (or influence function) at the true nuisance functions g_0 and p_0 :

$$\psi_0(Z) = \underbrace{g_0(1, X) - g_0(0, X)}_{\text{G-formula part}} + \underbrace{a_0(D, X) \cdot (Y - g_0(D, X))}_{\text{IPW correction part}}$$

where $a_0(D, X) = \frac{D}{p_0(X)} - \frac{1-D}{1-p_0(X)}$. It is sometimes convenient to write this weighting term as the residual of the treatment (the part of the treatment assignment that is not predictable from covariates) divided by the conditional variance of the treatment, i.e.,

$$a_0(D, X) = \frac{D - p_0(X)}{p_0(X)(1 - p_0(X))} \quad (1)$$

The asymptotic variance of the DR estimator is $\frac{1}{n} \text{Var}(\psi_0(Z))$. Let's compute $\text{Var}(\psi_0(Z)) = \mathbb{E}[(\psi_0 - \text{ATE})^2]$. We can expand the variance as follows:

$$\begin{aligned} \text{Var}(\psi_0) &= \underbrace{\mathbb{E}[(g_0(1, X) - g_0(0, X) - \text{ATE})^2]}_{\text{Term A: Variance of CATE}} \\ &\quad + \underbrace{\mathbb{E}[a_0(D, X)^2 (Y - g_0(D, X))^2]}_{\text{Term B: Variance of Weighted Residuals}} \\ &\quad + \underbrace{2\mathbb{E}[(g_0(1, X) - g_0(0, X) - \text{ATE}) \cdot a_0(D, X)(Y - g_0(D, X))]}_{\text{Term C: Cross-term}} \end{aligned}$$

Step 1: The Cross-Term Vanishes A key simplification occurs because the cross-term (Term C) is zero. This follows from the law of iterated expectations and the definition of the CEF:

$$\begin{aligned} C &= 2\mathbb{E}\left[(g_0(1, X) - g_0(0, X) - \text{ATE}) \cdot a_0(D, X) \cdot \mathbb{E}[Y - g_0(D, X) \mid D, X]\right] \\ &= 2\mathbb{E}\left[(g_0(1, X) - g_0(0, X) - \text{ATE}) \cdot a_0(D, X) \cdot 0\right] \\ &= 0 \end{aligned}$$

This is because $\mathbb{E}[Y \mid D, X] = g_0(D, X)$ by definition of the true conditional expectation function.

Step 2: Simplifying the Remaining Terms Now we simplify Terms A and B. **Term A** is the variance of the Conditional Average Treatment Effect (CATE) around the ATE:

$$A = \mathbb{E}[(\text{CATE}(X) - \text{ATE})^2] = \text{Var}(\text{CATE}(X))$$

where $\text{CATE}(X) = g_0(1, X) - g_0(0, X)$. **Term B** is the expected weighted variance of the outcome noise. Let $\epsilon = Y - g_0(D, X)$ be the residual noise, and let $\sigma^2(d, X) = \text{Var}(Y \mid D = d, X)$ be the conditional variance of the outcome (which can be heteroskedastic). By the law of iterated

expectations:

$$B = \mathbb{E}[a_0(D, X)^2 \cdot \mathbb{E}[\epsilon^2 \mid D, X]] = \mathbb{E}[a_0(D, X)^2 \cdot \sigma^2(D, X)]$$

Expanding $a_0(D, X)^2$ and using $\mathbb{E}[D \mid X] = p_0(X)$:

$$\begin{aligned} B &= \mathbb{E} \left[\frac{D}{p_0(X)^2} \sigma^2(1, X) + \frac{1-D}{(1-p_0(X))^2} \sigma^2(0, X) \right] \\ &= \mathbb{E} \left[\frac{\sigma^2(1, X)}{p_0(X)} + \frac{\sigma^2(0, X)}{1-p_0(X)} \right] \end{aligned}$$

The Final Variance Formula This leads us to the final variance formula.

The Semiparametric Efficiency Bound

$$\text{Var}(\psi_0) = \underbrace{\text{Var}(\text{CATE}(X))}_{\text{CATE Heterogeneity}} + \underbrace{\mathbb{E} \left[\frac{\sigma^2(1, X)}{p_0(X)} + \frac{\sigma^2(0, X)}{1-p_0(X)} \right]}_{\text{Weighted Noise Variance}}$$

4.2 Interpreting the Variance Components

Component 1: $\text{Var}(\text{CATE}(X))$ This component captures how much the treatment effect varies across individuals based on their characteristics. More heterogeneous effects lead to higher variance. This is intuitive: if the treatment helps some people a lot and hurts others, the average effect is harder to pin down because of this underlying variability.

Component 2: $\mathbb{E} \left[\frac{\sigma^2(1, X)}{p_0(X)} + \frac{\sigma^2(0, X)}{1-p_0(X)} \right]$ This component captures how noisy the unpredictable part of the outcome is, weighted by the inverse propensity. This phenomenon, where the variance of the error term ϵ depends on the covariates and treatment status, is what is known in classical statistics as **heteroscedasticity**.

- When $p_0(X) \approx 0$: we care a lot about $\sigma^2(1, X)$ (few treated).
- When $p_0(X) \approx 1$: we care a lot about $\sigma^2(0, X)$ (few controls).
- **Worst case:** The variance will be highest if we tend to not treat individuals for whom the outcome under treatment is very noisy (i.e., $p_0(X)$ is small when $\sigma^2(1, X)$ is large), and simultaneously tend to treat individuals for whom the outcome under control is very noisy (i.e., $p_0(X)$ is large when $\sigma^2(0, X)$ is large).

Intuitive Examples of Heteroscedasticity. Consider a medical trial for a new drug. If patients with a severe prior condition have very predictable outcomes (e.g., they almost certainly do not survive without treatment), then $\sigma^2(0, X)$ is small for this group. However, under the new treatment, some may survive and some may not, leading to a high $\sigma^2(1, X)$. If we tend to give the treatment to the sickest patients (high $p_0(X)$), we are in a favorable situation for the second term. But if we tend to withhold treatment from the sickest patients (low $p_0(X)$), we pay a large penalty for the high $\sigma^2(1, X)$ in that group.

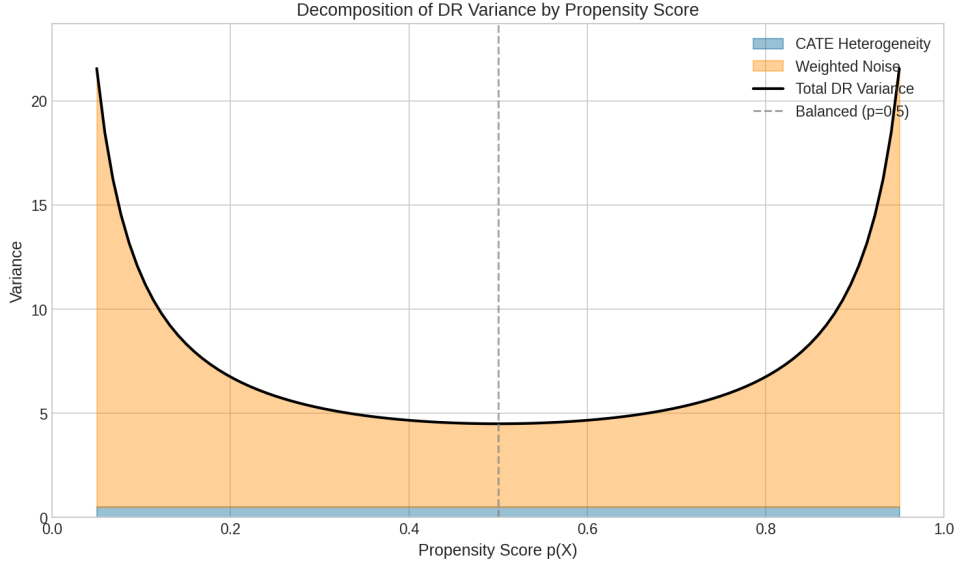


Figure 1: Decomposition of the DR estimator’s variance. The total variance (black line) is the sum of variance from CATE heterogeneity (blue area) and the weighted noise variance (orange area). The noise component dominates and explodes as the propensity score $p(X)$ approaches 0 or 1. *DGP: This is a theoretical plot where we assume homoskedastic noise $\sigma^2(1, X) = \sigma^2(0, X) = 1$ and fixed CATE variance $\text{Var}(\text{CATE}(X)) = 0.5$. We then plot the variance formula for different values of $p(X)$.* To replicate the plot examine this Notebook.

5 Application to Randomized Controlled Trials

We now turn back to the world of RCTs and examine the problem of precision. How and why should we use available pre-treatment variables X in an experiment?

Main Idea

Use covariates to **reduce variance** by explaining away predictable variation in the outcome.

Example: In a medical trial, patients with prior conditions will have worse survival. In an online platform, users with high prior engagement will likely engage more in the future. Prior covariates strongly predict future outcomes. The core idea is to use them to first predict a baseline response, and then measure the treatment effect on top of that baseline, thereby explaining away predictable variation and only paying for the residual, unpredictable noise.

5.1 Two-Means Estimator Variance

Consider an experiment where we treat with probability q . The simple two-means estimator is:

$$\hat{\theta}_{\text{TM}} = \bar{Y}_1 - \bar{Y}_0$$

Its variance is:

$$\text{Var}(\hat{\theta}_{\text{TM}}) = \frac{\text{Var}(Y \mid D = 1)}{q} + \frac{\text{Var}(Y \mid D = 0)}{1 - q}$$

Problem: With this estimator, we pay for the *entire* variance of the outcome Y within each treatment arm, even the portion that is systematically predictable from baseline covariates X .

The Ideal: Residual Variance Ideally, we would want to only pay:

$$\frac{\text{Var}(Y - g_0(1, X) \mid D = 1)}{q} + \frac{\text{Var}(Y - g_0(0, X) \mid D = 0)}{1 - q}$$

where $g_0(D, X) = \mathbb{E}[Y \mid D, X]$. This is the variance of the *residuals* after removing the predictable part.

Recall: The DR variance in an RCT (where $p(X) = q$) simplifies to:

$$\text{Var}(\text{CATE}(X)) + \frac{\mathbb{E}[\sigma^2(1, X)]}{q} + \frac{\mathbb{E}[\sigma^2(0, X)]}{1 - q}$$

Note that $\mathbb{E}[\sigma^2(d, X)] = \mathbb{E}[(Y - g_0(d, X))^2 \mid D = d]$. So the DR estimator gets us very close to this ideal variance, i.e., it achieves:

$$\text{Var}(\text{CATE}(X)) + \frac{\text{Var}(Y - g_0(1, X) \mid D = 1)}{q} + \frac{\text{Var}(Y - g_0(0, X) \mid D = 0)}{1 - q}$$

The only extra term is the variance of the CATE, which typically only contributes a tiny amount to the overall variance. Moreover, as we've already seen, this variance is unavoidable, since the DR estimator achieves the best variance of any possible estimator.

5.2 DR in RCTs: All Assumptions Can Be Dropped

Well, one might complain that the DR achieves this lower variance, but requires so many assumptions for this variance to be valid and so that it is asymptotically normal. However, in an RCT the story simplifies nicely. In an RCT with treatment probability q :

- The propensity score is **known and constant**: $p(X) = q$.
- The Horvitz-Thompson weights simplify:

$$a(D) = \frac{D}{q} - \frac{1 - D}{1 - q} = \frac{D - q}{q(1 - q)}$$

Note: This no longer depends on X !

- The DR formula becomes:

$$\hat{\theta}_{\text{DR}} = \mathbb{E}_n[\hat{g}(1, X) - \hat{g}(0, X)] + \mathbb{E}_n \left[\frac{D - q}{q(1 - q)} (Y - \hat{g}(D, X)) \right]$$

What about our convergence conditions?

- **Product rate condition:** $\sqrt{n} \cdot \text{RMSE}(\hat{g}) \cdot \text{RMSE}(\hat{p}) \rightarrow 0$
 - Automatically satisfied since $\text{RMSE}(\hat{p}) = 0$ (we know p exactly).

- **Propensity consistency:** $\hat{p} \rightarrow p_0$
 - Trivially satisfied since $\hat{p} = q = p_0$.
- **Outcome regression consistency:** $\hat{g} \rightarrow g_0$?
 - Surprisingly, this is **not needed**. A more refined theorem for the RCT setting shows that as long as our outcome regression model $\hat{g}$ converges to *any* limit function g_* as the sample size grows, the DR estimator remains consistent and asymptotically normal. The limit g_* does not have to be the true CEF g_0 . We only need $\hat{g} \rightarrow g_*$ for *some* limit g_* .

A Surprising Result: No Consistency Needed

In an RCT, we only need $\hat{g}$ to converge to *some* limit g_* , not necessarily the true CEF g_0 . Without any other assumptions (other than benign regularity assumptions) DR is asymptotically normal and centered at the true ATE:

$$\sqrt{n}(\hat{\theta} - \text{ATE}) \overset{a}{\sim} N(0, V(g_*))$$

with variance:

$$V(g_*) := \text{Var}(g_*(1, X) - g_*(0, X) + a(D)(Y - g_*(D, X)))$$

Examples Many outcome regression approaches will satisfy this convergence to a limit property.

- Linear regression over low-dimensional features: converges to the population best linear predictor (minimizing RMSE over all linear functions).
- Lasso over high-dimensional features: under sparsity, also converges to the best linear predictor.

What is not lost

It is important to clarify what is *not* lost in this setting:

- **Unbiasedness:** The estimator is still asymptotically unbiased and centered at the true ATE.
- **Root-n convergence:** The estimator still converges at the parametric $\sqrt{n}$ rate.
- **Valid confidence intervals:** The asymptotic normality result still holds, so confidence intervals and p-values remain valid.

The Catch!

What is lost: The limit variance $V(g_*)$ is not the semiparametric efficient variance $V(g_0)$. If g_* is, for example, the best linear predictor instead of the true CEF, then $V(g_*)$ will only account for the *linearly* predictable part of the outcome variance, not everything that is explainable. This means $V(g_*) \geq V(g_0)$.

Key Question (Next Lecture): The limit variance of the DR when the outcome regression converges to a limit $g_* \neq g_0$ can be larger than the best achievable variance. But is it even guaranteed that it is better than the variance of the simple two-means estimate, no matter what g_* is? We will address this question in the next lecture.

6 Summary: Four Key Takeaways

1. The DR estimator achieves the **semiparametric efficiency bound**—it’s optimally precise in observational settings.
2. DR variance depends on **CATE heterogeneity** and **noise amplified by extreme propensities**.
3. In RCTs, regression adjustment can **reduce variance**.
4. In RCTs, DR estimator is consistent and asymptotically normal for the ATE, even if outcome regression does not converge to the CEF but to some other limit g_* .

7 Next Time

- Examine precision improvement in RCTs via linear adjustment
- Revisiting and gaining a deeper understanding of OLS from a different perspective
- Continuous Treatments. Partially linear models and another “insensitive” formula.

8 References

- Zhong, Y., et al. (2022). Use of machine learning to estimate the per-protocol effect of low-dose aspirin on pregnancy outcomes. *JAMA Network Open*, 5(3), e2143414.

A Appendix: In-Class Activity - Estimator Pros and Cons

Estimator		Pros	Cons
Linear Regression Adjustment		Simplicity; No hyperparameter tuning (reduces p-hacking); Interpretable (even to non-ML reviewers); Provides standard confidence intervals; Highly precise if CEF is truly linear.	Likely misspecified; Assumes linear CEF; Can extrapolate significantly and provide misleading estimates if overlap is poor.
G-formula learner)	(S-	Can capture non-linearities by fitting a single model.	Slow convergence rate; Can be overconfident (overfitting) if not cross-fit; No straightforward CIs; May regularize away a small but real treatment effect to optimize overall MSE.
G-formula learner)	(T-	Often has better bias properties than S-learner because it cannot regularize away the treatment effect (fits separate models).	Can have poor accuracy for the minority treatment group in imbalanced datasets (e.g., 5% treated); Does not share information between treatment arms; Can exacerbate extrapolation issues with weak overlap.
IPW (ML)		Only requires learning the propensity score, not the outcome model; Can handle underrepresented subgroups well if propensity is modeled correctly.	High variance; Very sensitive to poor overlap (explodes with extreme propensity scores); No straightforward CIs with ML models due to first-order sensitivity.
IPW (Logistic)		Simpler than ML-based IPW; Provides confidence intervals (e.g., via the delta method).	Assumes a simple (logistic) form for the propensity score model, which is likely misspecified.
Doubly (DR)	Robust	Double robustness; Lower variance than IPW; Asymptotically normal (root-n convergent) even with ML, which allows for valid CIs; Insensitive to ML errors (Neyman Orthogonality); First ML-based method with confidence intervals.	Cannot empirically test the product rate condition for convergence; More randomness due to k-fold cross-validation (can be stabilized by drawing many folds and taking the median of estimates and standard errors); Still suffers from high variance with poor overlap.
DR (Semi-crossfitting)	(Semi-	Better computational performance; Statistically better for hyperparameter tuning.	Fewer guardrails against overfitting; Susceptible to bias if tuning is too extensive relative to sample size.
DR (Stacked-semi-crossfitting)		Reduces variance of tuning; Can improve RMSE of nuisance models through ensembling, helping satisfy the product rate condition.	Even fewer guardrails against overfitting; Must be careful to ensemble only a small set of models to avoid overfitting the stacking weights.